

Every workload likes fast cache memory banks but when you look at gaming workloads then there is a ridiculous number of simultaneous instructions piling up which is easier to handle when you have large caches, especially when there are a large number of cores. In the case of AMD, it's not even the flagship processor that's leading in gaming benchmarks. The Ryzen 7 7800X3D which is a mid-tier processor is better than the more expensive flagships. This is because games can only make optimal use of 4-8 cores and more isn't always better. That's not to say that Intel isn't good. In 1080p, 1440p and 4K gaming benchmarks, the Ryzen 7 7800X3D is only ahead by 1-5 per cent. The performance gap is widest at 1080p and narrows down to 1 per cent at 4K. So you can go with either Intel or AMD for your gaming needs and you'll do great. There are very few games that give double-digit performance differences. The Ryzen 7 7800X3D becomes the obvious choice simply because you get better performance at a little over half the price of the 14900K.

CONTENT CREATION

One of the fastest-growing use cases is that of content creation with everyone becoming a creator thanks to the advent of easy-to-use creator tools. Intel and AMD have a different family of processors for such needs. Intel used to have a consumer-grade HEDT platform but now it's mostly the Xeon family of processors that are used for heavy-duty content creation needs. AMD has had their Threadripper family targetted at similar workloads which typically run on workstations. AMD has the leadership here thanks to a ridiculously high core count. Their Threadripper 7995WX Pro with 96 cores can make mincemeat of any content creation workload that any creator can throw at it. In contrast, Intel has the Sapphire Rapids Xeon w9-3495X with 56 cores which lags behind the Threadripper flagship in Blender, V-Ray, Cinebench, Unreal, DaVinci Resolve, Premiere Pro and After Effects.



The Intel Core i9-14900K actually beats all the super-expensive workstation processors in Photoshop. We say super expensive because the AMD Threadripper 7995WX costs a whopping USD 9,999 and the Intel Xeon X9-3495X costs USD 5,889. On the cheaper consumer side of things, the Intel Core i9-14900K is a great processor for content creation that beats the competing flagship in most benchmarks.

POWER EFFICIENCY

AMD which was known for the longest time to be a power hog turned the tables with their current generation of processors. The Intel 14th Gen flagship has been guzzling well over 200 Watts of power on some of the more intensive content creation benchmarks. The AMD 7000 series processors including the flagship consume way less. The 14900K has a listed max Turbo power consumption of 253 Watts and in some workloads, it goes well beyond that. The AMD Ryzen 9 7950X also consumes a lot of power in similar workloads but not as much. Just to put things into perspective, the 7950X has consumed about 330 Watts in intensive multicore workloads and the Intel 14900K has managed to go up to 390 Watts. These are ridiculous numbers for flagships but that's what both brands are resorting to to get a little extra performance. From a power efficiency perspective, AMD clearly has the advantage.



VALUE FOR MONEY

This is fairly easy to discern once you have read through all the previous parameters. AMD's processors have the edge in a lot of intensive workloads such as gaming but when you look at day-to-day use cases, the two are extremely competitive. On the workstation side, AMD processors beat Intel's workstation offerings but they also cost a lot more. On the consumer side, it's Intel that stays ahead. Then there's the power efficiency metric wherein AMD becomes the winner. AMD's flagship 7950X3D and Intel's flagship 14900K cost about the same at INR 55K so you can opt for either and have a great desktop experience. For gaming, you can save some money with 7800X3D since it costs about 36K and invest the money saved in a more powerful graphics card.

WHAT'S THE WORD, DOC?

With the way things are progressing. There's no telling if the PC ecosystem will continue to be a two-player market. There are new players on the horizon and the current players have done away with the typical 2-year cadence to ship products as and when they are ready. With companies such as Qualcomm throwing their hat into the ring, we might see the ecosystem completely disrupted. Until then, you get to enjoy this great period wherein every manufacturer is trying its best to put out the most powerful product to market earliest. **[d]**